

Problem

Which of the following is not a right way to express the sinusoid $A \cos \omega t$?

- (a) $A \cos 2\pi ft$
- (b) $A \cos(2\pi t/T)$
- (c) $A \cos \omega(t - T)$
- (d) $A \sin(\omega t - 90^\circ)$

Step-by-step solution

Step 1 of 4

Consider the following sinusoid:

$$f(t) = A \cos \omega t$$

Consider the following function:

$$\begin{aligned} f'(t) &= A \cos(2\pi ft) \\ &= A \cos \omega t \end{aligned}$$

Here, ω is the radial frequency.

$$\omega = 2\pi f \text{ rad/s}$$

Therefore, $A \cos(2\pi ft)$ is a right way to express the sinusoid $A \cos \omega t$.

Hence, option (a) is incorrect.

Step 2 of 4

Consider the following function:

$$\begin{aligned} f'(t) &= A \cos\left(\frac{2\pi t}{T}\right) \\ &= A \cos \omega t \end{aligned}$$

Where,

$$\omega = \frac{2\pi}{T} \text{ rad/s}$$

Therefore, $A \cos\left(\frac{2\pi t}{T}\right)$ is a right way to express the sinusoid $A \cos \omega t$.

Hence, option (b) is incorrect.

Step 3 of 4

Consider the following function:

$$\begin{aligned}f'(t) &= A \cos \omega(t-T) \\&= A \cos \frac{2\pi}{T}(t-T) \\&= A \cos \left(\frac{2\pi t}{T} - 2\pi \right) \\&= A \cos \left(\frac{2\pi t}{T} \right)\end{aligned}$$

$$= A \cos \omega t$$

Where,

$$\omega = \frac{2\pi}{T} \text{ rad/s}$$

Therefore, $A \cos \omega(t-T)$ is a right way to express the sinusoid $A \cos \omega t$.

Hence, option (c) is incorrect.

Step 4 of 4

Consider the following function:

$$\begin{aligned}f'(t) &= A \sin(\omega t - 90^\circ) \\&= -A \cos \omega t\end{aligned}$$

Therefore, $A \sin(\omega t - 90^\circ)$ is not a right way to express the sinusoid $A \cos \omega t$.

Hence, the correct option is (d).